

VAE

It is an elegant
Tea Drinking, Probabi-
-listic lady.

GAN

It is a Gambler,
who loves a high
stake game of
MIN-MAX

GAN

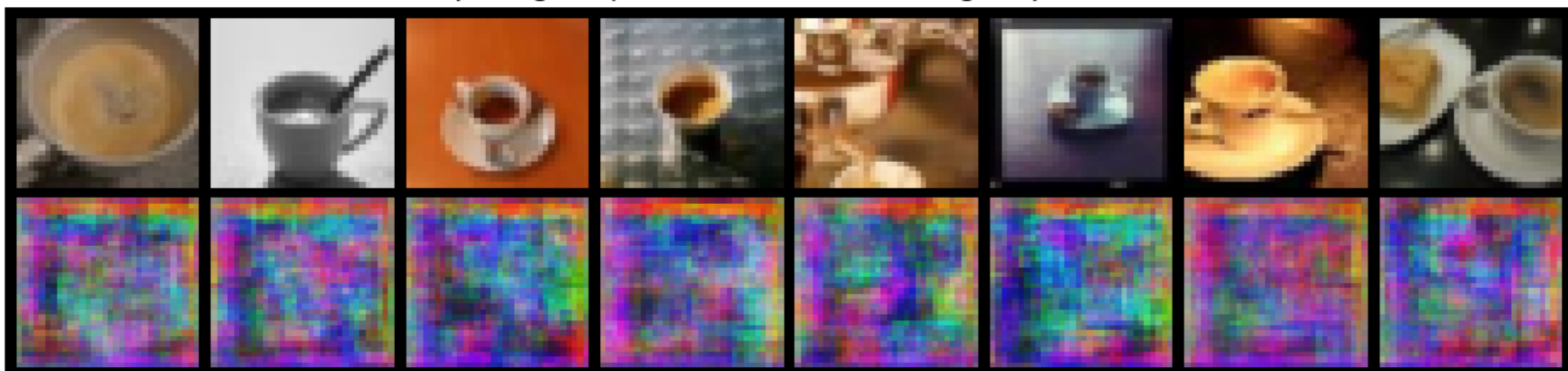
Forger (Generator)

Never seen a cup of
coffee & somehow
creates a forgery.

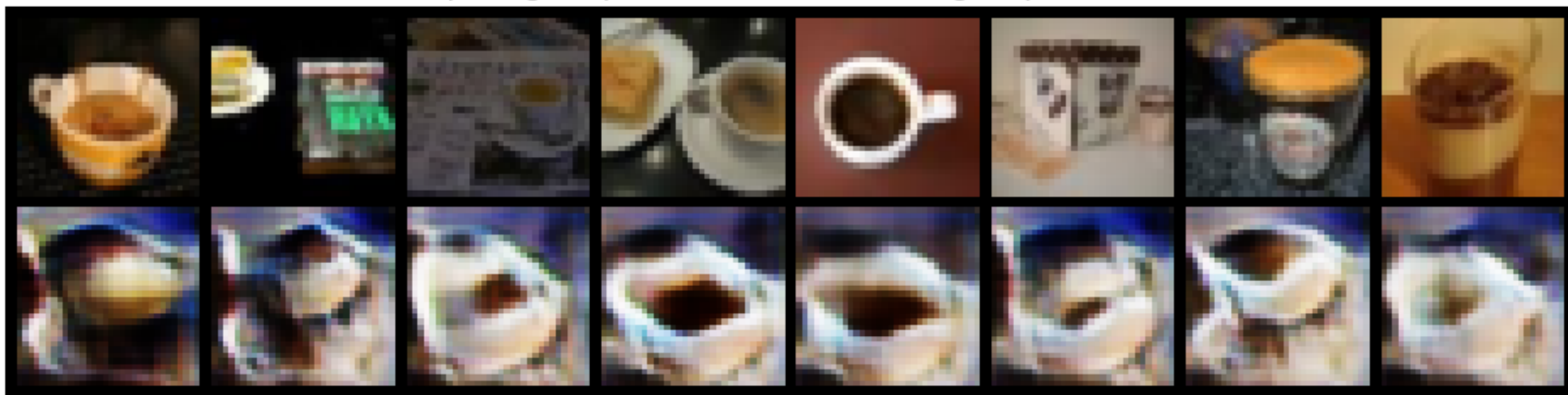
Detective (Discriminator)

Job is to look at
coffee images & forgery
to differentiate b/w the
two.

Top: Original | Bottom: Generated Image (Epoch 10)



Top: Original | Bottom: Generated Image (Epoch 990)



The cross checking of Forgery & Creation of better Forgery is an
ADVERSARIAL GAME

$$Z \sim \mathcal{N}(0, I)$$



$$G(Z) = \hat{x}$$

P_{data}

x



$D(x)$

$\hat{x}$

$$D(\hat{x}) = D(G(Z))$$

What is $P_{data} > P_g > P_z$?

$p_{data}(x) \rightarrow$ If I close my eyes and pick a random image out of True dataset what is the prob. of picking image x .

$P_g(x) \rightarrow$ If I ask my generator to generate a random image, what is probability to create exact image x .

$\phi_z(z) \rightarrow$ If I reach into a mathematical Shape (Gaussian Curve), what is probability of pulling out exact string of random numbers z .

KAPOW!!!

P_g, P_{data}, P_z is the Bag

Objective of Discriminator:

$D(x) \rightarrow 1$ (Real is Real)

$D(G(z)) \rightarrow 0$ (Fake is Fake)

Case-I $x \in P_{\text{data}}$

$$L_d^D = \min_D (-\log(D(x)))$$

$$= \max_D (\log(D(x)))$$

Case-II $x \in P_g$

$$L_g^D = \min_D (-\log(1 - D(G(z))))$$

$$= \max_D (\log(1 - D(G(z))))$$

$$L = L_d^D + L_g^D$$

$$= \max_D \{ \log(D(x)) + \log(1 - D(G(z))) \}$$

For Generator $\rightarrow$

$$D(G(z)) \rightarrow 1$$

$$D(x) \rightarrow G \text{ cannot influence } D$$

But why??

Case - I : Try and fool D with $G(z)$

$$L_g^G = \min_G \{ \log D(G(z)) \}$$

$$= \max_G \{ \log D(G(z)) \}$$

$$\Rightarrow \min_G \{ \log(1 - D(G(z))) \}$$

Instead of maximizing its own success, the Generator works to undermine/minimize the Decoder's success.

BOOM!

$$\min_G \max_D \left\{ \log(D(x)) + \log(1 - D(G(z))) \right\}$$

Generator is NEVER! EVER!
shown P_d .

Initially the generator produces static noise and it becomes easy for D to discriminate P_{data} & P_{True} .

$$D(G(z)) \rightarrow 0$$

$$\min_G [\log(1 - D(G(z)))]$$

$$\min_G [\log(1)] = 0$$

Gradient saturation is reached

So we use $\min_G [-\log D(G(z))]$ instead.

Nash Equilibrium

Unlike NNs where minimization is the goal, GANs are formulated as a non-cooperative MIN-MAX game.

And objective is achieving Nash Equilibrium, a state where neither the Generator nor Discriminator can improve their position given the parameters.

Good ol' fashioned Stalemate.

Model Expressivity

If D is much more expressive than G , G gets zero gradients & stalls.

If G is more expressive than D , G will find a bizarre, high-frequency mathematical loop-hole that fools D .